A. Mr. Kitayuta's Gift

time limit per test 1 second
memory limit per test 256 megabytes
input stdin
output stdout

Rating: 1100 | Tags: brute force, implementation, strings | https://codeforces.com/problemset/problem/505/A

Mr. Kitayuta has kindly given you a string *s* consisting of lowercase English letters. You are asked to insert exactly one lowercase English letter into *s* to make it a palindrome. A *palindrome* is a string that reads the same forward and backward. For example, "`noon`", "`testset`" and "`a`" are all palindromes, while "`test`" and "`kitayuta`" are not.

You can choose any lowercase English letter, and insert it to any position of *s*, possibly to the beginning or the end of *s*. You have to insert a letter even if the given string is already a palindrome.

If it is possible to insert one lowercase English letter into *s* so that the resulting string will be a palindrome, print the string after the insertion. Otherwise, print "`NA`" (without quotes, case-sensitive). In case there is more than one palindrome that can be obtained, you are allowed to print any of them.

Input

The only line of the input contains a string *s* ($1 \le |s| \le 10$). Each character in *s* is a lowercase English letter.

Output

If it is possible to turn *s* into a palindrome by inserting one lowercase English letter, print the resulting string in a single line. Otherwise, print "`NA`" (without quotes, case-sensitive). In case there is more than one solution, any of them will be accepted.

Examples

Input
`revive`

Output
`reviver`

Input
`ee`

Output
`eye`

Input
`kitayuta`

Output
`NA`

Note

For the first sample, insert '`r`' to the end of "`revive`" to obtain a palindrome "`reviver`".

For the second sample, there is more than one solution. For example, "`eve`" will also be accepted.

For the third sample, it is not possible to turn "`kitayuta`" into a palindrome by just inserting one letter.